



Stream power

NVIDIA shows interest in using CUDA with antivirus software

YouTube, the popular online video hosting service gets a billion views a day

Graphic Cards Test

MADE FOR EACH OTHER

Matching your display size with your graphics card

So you just spent Rs. 15,000 on a new graphic card and you're burning to see how good *Crysis Warhead* looks on a large screen. So you buy a 24-inch LCD monitor. Upon running the game at a glorious resolution of 1920 x 1200 pixels at ultra settings you suddenly realise your game is framing and in-game gunfights sequences appear to run in slow motion. You, sir, are a victim of 'improper-display-matching' as we like to call it. The fact is when choosing a monitor for gaming you need to be doubly careful about the kind of games you're going to be running. There's no sense in buying a super large monitor only to find the games you play slowing to a crawl and having to adjust the resolution and/or eye-candy down as it'll ruin the experience. Unless you've got pots of money to blow on SLI and Crossfire configurations involving high-end cards, pay close attention to which games you play and how fast they run on your current monitor and resolution. For people looking to play demanding games like *Warhead*, *Clear Sky*, *GRID 2* and so on here are a few formulae to help you: generally, for an NVIDIA GTX 260 / ATI HD 4870 class card, we recommend a 19 to 22-inch display, nothing larger. This equals a resolution of 1680 x 1050 pixels. Those with NVIDIA GTX 285 / ATI HD 4890 and above cards can consider a 24-inch display (equating to 1920 x 1200 pixels). If you have anything in the GeForce 9800 GTX / Radeon HD 4850 range of cards forget about anything larger than a 19-inch LCD with a maximum resolution of 1440 x 900 pixels. This will ensure your games run at playable fps, although to be honest this is also dependent on the games you play.

MSI	ASUS	ASUS	ASUS	ASUS
N9800GTX PLUS-T2D512-OC	EN 9800 GT MATRIX	EAH 4850 TOP	EAH 4850 MATRIX	EN 9800GTX DK TOP
GeForce 9800GTX	GeForce 9800GT	Radeon HD 4850	Radeon HD 4850	GeForce 9800GTX
Rs. 11,000	Rs. 13,000	Rs. 14,950	Rs. 15,700	Rs. 15,700
63.51	51.47	58.94	58.74	62.48
5.85	5.48	6.23	5.85	6.60
57.66	45.99	52.71	52.89	55.88
128	112	160 (800 / 5)	160 (800 / 5)	128
760 / 1880	600 / 1500	680 / -	625 / -	775 / 1950
GDDR3 / 2300	GDDR3 / 2600	GDDR3 / 2100	GDDR3 / 1986	GDDR3 / 2360
512 / 256-bit	512 / 256-bit	512 / 256-bit	512 / 256-bit	512 / 256-bit
2 x DVI, 1 x S-Video	1 x DVI, 1 x S-Video, 1 x HDMI	2 x DVI, 1 x S-Video	2 x DVI, 1 x S-Video	2 x DVI, 1 x S-Video
1 x DVI-D-Sub converter, Crossfire connect, S-Video-Composite connect, S-Video to component	1 x DVI - D-Sub converter, 1 x DVI - HDMI converter, HDMI audio cable, 1 x 6 pin to 4 pin molex power converter	1 x DVI - D-Sub converter, 1 x DVI - HDMI converter, HDMI audio cable, 1 x 6 pin to 4 pin molex power converter	1 x DVI - D-Sub converter, 1 x DVI - HDMI converter, HDMI audio cable, 1 x 6 pin to 4 pin molex power converter	1 x DVI - D-Sub converter, 1 x DVI - HDMI converter, HDMI audio cable, 1 x 6 pin to 4 pin molex power converter
-	-	-	-	-
41 / 62	52 / 65	45 / 51	43 / 55	52 / 60
8798 / 6936	6221 / 4789	9092 / 7655	8499 / 7098	9041 / 7137
23 / 16 / 10	18 / 12 / 6	20 / 14 / 6	20 / 14 / 9	24 / 16 / 10
30 / 19 / 12	23 / 15 / 8	25 / 18 / 12	24 / 17 / 11	31 / 20 / 12
45.1 / 18 / 7.4	23.5 / 10 / 4.8	30.5 / 16.5 / 9	28.5 / 15.5 / 9.1	32.4 / 13.2 / 5.5
46.1 / 18.1 / 8.7	31.4 / 13.4 / 6.4	42.4 / 21.1 / 7.5	39.7 / 20.1 / 8.3	46.7 / 18.4 / 7
58.37 / 38.2 / 16.23	46.03 / 28.23 / 14.88	41.75 / 17.97 / 4.04	47.22 / 30.44 / 19.01	55.6 / 37.14 / 12.75
67.4 / 42.92 / 20.19	50.61 / 36.24 / 22.37	163.69 / 39.87 / 28.06	51.1 / 26.86 / 15.8	93.01 / 44.07 / 20.32
130 / 163	99 / 122	114 / 143	108 / 136	132 / 155
289 / 138.3 / 65.2	239.2 / 104.8 / 47.9	197 / 108.5 / 50.7	253 / 140.9 / 70.7	319.6 / 143.9 / 64.3
425.5 / 190.1 / 93	326.6 / 146.2 / 76.7	250 / 148.8 / 72.2	196 / 103 / 44.9	420 / 199.8 / 100.8
8 / 7.75	8 / 7.75	8 / 8	8 / 8	8 / 7.75

an extravagance by some there is actually good value-for-money to be had. Sounds shocking doesn't it? People build entire PCs for Rs. 15,000 these days, but we're not talking about *those* PCs. We're talking about gaming rigs replete with quad core processors, 4 GB of RAM and RAID 0 storage etc. ATI's Radeon HD 4870 and newer 4890 clash with NVIDIA's GTX 260 and GTX 275 GPUs. And a mighty battle it is, with contestants from both sides drawing blood. One thing needs clearing, NVIDIA officially has two versions of the GTX 260 and the naming convention

used to distinguish between the two is horrible. The older, (and supposedly phased out), GPU, is called the GTX 260 and has 192 stream processors. The newer one has 216 stream processors and is called the GTX 260 216. Some people like ZOTAC choose to call theirs the GTX 260² while others like ASUS and XFX do not really differentiate and the only way we could tell the

difference was using GPU-Z. Although the newer 216 SP GPU is faster, an overclocked GTX 260 can easily perform comparably. ATI's



Palit GTX 260 SONIC 216

Radeon HD 4870 and 4890 are also nearly identical GPUs. AMD chose to keep the same number of clusters of SPs, so both have 800 units. However, the clocks on the HD 4890 are higher, evidently with the 4870 over-clocking really well, it was only a matter of time before AMD tweaked things. With the core running at a whopping 850 MHz, the HD 4890 meant to take on the GTX 275, while the HD 4870 is positioned against the GTX 260 216.

Unlike the HD 4850 that had a pretty useless single-slot cooling solution the HD 4870